



Louie Jason

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Electrical Engineering student with hands-on experience in robotics systems, IoT, and software-hardware integration. Experienced in working with Raspberry Pi in Linux environments, MQTT-based communication systems, and robotics development. Skilled in Python, debugging, troubleshooting, and system performance analysis. Strong interest in robotics operations, automation systems, and software reliability.

EDUCATION

Krida Wacana Christian University, Jakarta

Sep 2022 – Present

Undergraduate, Electrical Engineering

- GPA 3.69 of 4.00
- 2nd Highest GPA – Semesters I–III (2022-2023)

SKILLS

Programming & Scripting: Python, SQL, C++, Arduino Programming

Systems & Tools: Raspberry Pi (Linux), MQTT, Git, Jupyter Notebook, VS Code

Robotics & System Engineering: Troubleshooting & Debugging, System Integration, Hardware-Software Integration, Robotics Systems, IoT Systems

Data & Analytics: Data Analysis, Pandas, NumPy, Microsoft Excel, System Performance Analysis

PROJECTS

Undergraduate Thesis – Smart Chair Posture Classification System

- Developed an IoT-based smart chair system for real-time posture monitoring using ESP32, FSR sensors, IMU sensors, Firebase Realtime Database, and Streamlit.
- Built a real-time data collection and labeling pipeline by transmitting sensor data from ESP32 to Firebase and processing it through a Streamlit dashboard for monitoring and prediction.
- Applied machine learning models including Random Forest, KNN, and XGBoost using Python for posture classification, achieving 98.03% accuracy with the final XGBoost model.

Robotics Development – KRSBI 2022 (@Institut Teknologi Sepuluh Nopember)

- Participated in robotics system development and integration for KRSBI competition at ITS.
- Worked on robot testing, troubleshooting, and hardware-software coordination in competitive robotics environments.
- Collaborated with team members to improve robot performance and operational reliability.

Smart Fire Notification System using Raspberry Pi & MQTT

- Developed an IoT-based fire notification system for multi-storey buildings using Raspberry Pi in a Linux environment and Python programming.
- Implemented MQTT communication for real-time data transmission and system integration without requiring extensive cabling, improving scalability and deployment efficiency.
- Performed debugging, system monitoring, troubleshooting, and iterative testing to optimize system reliability and operational performance.

WORKING EXPERIENCE

Galva Technologies, Jakarta

Jul 2025 – Sep 2025

Research and Development Intern

- Built and tested 3 hardware/software prototypes to support R&D innovation.
- Performed hardware troubleshooting and debugging, improving system reliability.
- Optimized prototype performance through iterative testing, reducing development time by 15%.

Lecturer Assistant - Robotics

- Assisted lecturer in teaching Robotics course and conducting lab sessions.
- Guided students in building and programming robots using C++ and Arduino.

ACHIEVEMENTS

Funding Awardee – Program Kreativitas Mahasiswa Karsa Cipta (PKM-KC) 2025, Jakarta Aug 2025

Secured national funding for SITCARE (IoT Smart Chair) as 1 of 1,590 selected teams from 33,000+ proposals.

1st Place – Robotic Sumo Competition, FIK FAIR 2025 – UPN Veteran Jakarta Sep 2025

Generasi Teknologi Scholarship Awardee, Jakarta Sep 2022

Recipient of a full 8-semester scholarship for Engineering students with outstanding academic performance.

2nd Highest GPA – Semesters I–III (2022-2023), Jakarta Sep 2022

Consistently ranked as the 2nd highest GPA holder across the first three semesters.

LEADERSHIP ACTIVITIES

Program Kreativitas Mahasiswa – Karsa Cipta (PKM-KC), Jakarta Aug 2025

Team Leader

- Led 4-member team to develop IoT Smart Chair with posture monitoring & cloud integration.
- Coordinated research, hardware/software development, and reporting for PIMNAS selection.

UKRIDA Mengajar at Sinar Kasih School, Bogor Mar - Jun 2025

Team Leader - Coding Tutor

- Led a small team in teaching coding sessions as part of the UKRIDA Mengajar (Kampus Mengajar) program.
- Organized lesson plans, delegated teaching responsibilities, and ensured learning objectives were achieved.

CERTIFICATIONS

- Coding Camp 2026 – DBS Foundation (Data Science Path, Ongoing)
- Data Science Bootcamp – Digica Academy
- Intro to Data Analytics – RevoU
- SQL & Data Visualization – Dicoding
- Python & Data Science – DQLab

ORGANIZATIONS

Duta UKRIDA, Jakarta Sep 2023 - Present

Ambassador

- Served under the Marketing, Public Relations, and Admissions unit to promote the university.
- Represented UKRIDA through school visits, presentations, and events to attract prospective students.

Himpunan Mahasiswa Elektro, Jakarta Jul 2023 - Jul 2024

Head of Public Relations

- Directed the Public Relations Division of the Electrical Engineering Student Association.
- Oversaw communication strategies, built partnerships, and managed media relations.